

Automated Vehicle Access Control System

Presentation Chapter Summaries & Panel Defense Cheat-Sheet

1. CHAPTER-BY-CHAPTER PRESENTATION SUMMARIES

Chapter 1: Introduction

Summary: In modern urban settings like Iligan City, manual vehicle gates are slow and highly prone to human error, resulting in bottleneck traffic. This project leverages Verilog Hardware Description Language to automate parking barrier operations via a high-speed Finite State Machine (FSM). The scope covers digital RTL design and behavioral simulation inside Xilinx Vivado; it explicitly excludes structural mechanical fabrication or physical motor torque sizing.

Key Presentation Point: Emphasize that the system smoothly controls 4 states (IDLE, OPENING, OPEN, CLOSING) and implements a high-priority obstruction loop designed for immediate safety reversal back to OPENING.

Chapter 2: Review of Related Literature (RLL)

Summary: Finite State Machines stand as the reliable industry standard for sequential automation, establishing predictable transitions based on verified sensor evaluations. While historical, basic digital memory design focuses on asynchronous components like the cross-coupled NOR SR Latch, modern FPGA systems utilize edge-triggered D-Flip Flops. Synchronous elements eradicate the timing hazards, race conditions, and hardware glitches inherent to unclocked latches.

Key Presentation Point: Highlight that prioritizing safety-critical sensor inputs to interrupt and override mechanical closing sequences is a proven, documented standard for infrastructure preservation.

Chapter 3: Methodology

Summary: The hardware architecture maps synthesizable behavioral Verilog descriptions onto physical, synchronized FPGA structures. This implementation bypasses glitch-prone asynchronous configurations by utilizing edge-triggered registers. During compilation, Vivado maps the code onto D-Flip Flops for synchronized state storage, Look-Up Tables (LUTs) for combinational gating expressions, and Multiplexers (MUXes) to route next-state data lines.

Key Presentation Point: Detail how clock-synchronized memory locks incoming sensor events (State Persistence) so brief signal pulses are not forgotten, while positioning the obstruction statement at the top of the logic block creates a combinational priority network that instantly commands the D-Flip Flops to reverse state.

Chapter 4: Results & Discussion

Summary: System execution was rigorously verified through deterministic test cases and behavioral wave simulation inside Vivado. Running on a standard 100MHz system clock, the registers process state logic

relative to a strict 10ns clock cycle heartbeat. Test boundary simulations verified standard operational routing as well as concurrent, conflicting input events.

Key Presentation Point: Point directly to Test Case 05 (Conflict Analysis). When the exit sensor commands a close but the obstruction loop simultaneously asserts a hazard, the priority combinational logic instantly kills the close command, forcing the D-Flip Flops to register the safe OPENING state value within a single 10ns window.

Chapter 5: Conclusion & Recommendations

Summary: The project successfully verified a robust, fail-safe vehicle gate control system using a synthesizable FSM with total logical integrity. Future recommended stages include downloading the Verilog source code onto real physical FPGA chip boards (e.g., Basys 3 or Nexys Artix-7) and performing Static Timing Analysis (STA) to evaluate physical signal setup and hold compliance.

Key Presentation Point: Conclude by recommending the integration of Pulse Width Modulation (PWM) hardware blocks to smoothly manage the physical motor speed and acceleration curve profiles of an actual structural gate arm deployment.

2. CORE CONCEPT DEEP-DIVE: THE D-FLIP FLOP

A **D-Flip Flop (Data Flip-Flop)** is a synchronous binary memory primitive that samples the logic state of its input line (D) strictly on the rising edge of a clock trigger pulse (`posedge clk`) and updates its storage output (Q) accordingly. Between these clock edges, the internal memory ignores any fluctuations on the input wires, locking the circuit state and isolating it from external noise.

In this project, your 4-state architecture is stored in a 2-bit register array utilizing D-Flip Flops. Driven by a 100MHz clock frequency typical of modern FPGA boards, a rising clock edge triggers precisely every **10 nanoseconds** ($T = 1 / 100 \text{ MHz} = 10 \text{ ns}$). This clock cycle represents the system's exact evaluation timeline. If a safety event or payment pulse occurs, the maximum latency for the D-Flip Flops to capture the input and change states is bounded to a single 10ns window, yielding near-instantaneous hardware execution.

3. STRATEGIC PANEL Q&A RESPONSES

Panel Question: Your report text mentions "SR Latches" in earlier chapters, but your implementation uses "D-Flip Flops". Why are there two different memory elements discussed?

Your Answer: "The SR Latch discussion in the early chapters establishes the fundamental, historical foundation of digital memory—illustrating how cross-coupled feedback loops retain data. However, for our actual FPGA implementation, asynchronous SR Latches are highly discouraged because they introduce timing glitches and race conditions. Therefore, our project implements synchronous, edge-triggered D-Flip Flops to guarantee that all state retention and transitions are perfectly locked to a stable global clock domain."

Panel Question: How did you ensure that the obstruction signal overrides the exit_sensor if they are triggered at the exact same time (Test Case 05)?

Your Answer: "This was achieved through code structure priority. Inside our Verilog sequential block, the conditional safety statement `if (obstruction)` was intentionally placed outside and completely above the standard `case(gate_state)` evaluation. In hardware synthesis, this layout forces Vivado to generate a combinational look-up table (LUT) priority network. This network overrides any standard sensor commands and immediately routes the `OPENING` state code directly into the D-Flip Flop memory registers for the next clock edge."

Panel Question: What are the Setup (t_{su}) and Hold (t_{h}) times mentioned in your datasheet appendix, and why do they matter?

Your Answer: "Setup time is the minimum duration an input signal must remain completely stable before the rising edge of the clock arrives, and hold time is how long it must remain stable after the edge passes. They are critical because if our external sensors change state exactly on the clock edge, the internal D-Flip Flops can enter a state of metastability—a glitch where the hardware output gets stuck between logic '0' and '1'. Designing our combinational paths to stay well within our 10ns clock period guarantees that our signals settle cleanly, eliminating metastability errors."